HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and built **intelligent agents** and **reusable enterprise AI capabilities** using Python, integrating with **AWS Bedrock** for LLM access and experimentation.
- Developed **prompt frameworks** and integrated **tool calling capabilities**, enabling AI agents to perform complex tasks like data retrieval and system interaction.
- Implemented **Retrieval-Augmented Generation (RAG) architectures**, connecting AI models to enterprise knowledge bases for context-aware responses and accurate information retrieval.
- Engineered **autonomous workflows** and **API-driven solutions** in Python to automate business processes across departments, reducing manual intervention by an estimated **30%**.
- Collaborated with business stakeholders to identify and prototype AI solutions for operational challenges, translating requirements into **working AI prototypes** that demonstrate measurable business value.
- Contributed to the development of **enterprise AI standards** and reusable templates, focusing on secure, governed approaches to AI tool usage and system connectivity.
- Containerized AI agent services using **Docker** and deployed them to **AWS ECS**, configuring basic resource limits and network policies for predictable scaling and isolated execution.

SKILLS

AI/ML – Large Language Models (LLMs), AI agents, Prompt engineering, Retrieval-Augmented Generation (RAG), Function calling, Tool integration, Model orchestration, Multi-agent collaboration, AI evaluation, Generative AI

Programming Languages – Python, Java, SQL

Cloud Platforms – AWS (Bedrock, EC2, S3, Lambda, CloudWatch, ECS)

Frameworks/Orchestration – LangChain, OpenAI, Anthropic Claude

Databases – PostgreSQL, MySQL, Oracle, Vector Databases

DevOps/Tools – Docker, Git, Jenkins, Jira, Confluence

PROJECTS

Enterprise Knowledge Assistant (HCL Technologies)

- Developed a Python-based AI agent designed to answer queries from internal teams by accessing and synthesizing information from a large corpus of technical documentation.
- Implemented a **RAG architecture** using **OpenSearch** as the vector database and **FastAPI** for serving model inference requests, achieving ~**85%** accuracy on a benchmark test set of internal FAQs.
- Integrated agent with a **REST API** to fetch real-time operational status from core enterprise systems, enabling proactive issue identification and response.

Automated Workflow Orchestrator (HCL Technologies)

- Built a Python application to automate routine data processing and reporting tasks for financial operations, replacing a manual, multi-step process.
- Designed and implemented **autonomous workflows** that parsed incoming transaction data, performed validation checks, and generated summary reports, reducing report generation time by **60%**.
- Containerized the application using **Docker** for consistent deployment across development and production environments managed via **Jenkins** pipelines.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)